

CHAEWON YOON

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RESEARCH INTERESTS

Reliable scientific machine learning; neural operators; numerical and physical fidelity; low-precision inference; uncertainty and evidence-based verification; generative and unsupervised learning; evidence-grounded multimodal inference; auditable human-AI systems.

EDUCATION

University of Wisconsin-Madison - B.S. in Data Science

Expected May 2028

Incoming transfer student, Fall 2026. Fall coursework: ASTRON 103 - The Evolving Universe; MATH 345 - Linear Algebra and Optimization; STAT 240 - Data Science Modeling I.

University of Wisconsin Oshkosh - Undergraduate Study

Sep. 2024-May 2026

Relevant coursework: Programming I-II; Data Structures; Discrete Mathematics; Applied Statistics; Probability & Statistics; Calculus I-II; credit-bearing Independent Study in Numerical Methods; credit-bearing Independent Study in Neural Methods for PDEs.

PUBLICATIONS & RESEARCH MANUSCRIPTS

Peer-Reviewed Publication

- Yoon, C. "EvidenceLens: A Declarative Evidence-State Interface for Human-AI Visual Inspection." **ACM UIST Adjunct 2026 (Poster)**. [DOI](#) | [GitHub](#)

Manuscripts Under Peer Review

- Yoon, C. "Conditional Precision Bottlenecks and Quantized Defect Correction in Fractional Neural Operators." Manuscript under peer review; revised preprint and public research artifact, 2026. [GitHub](#) | [Software DOI](#)
- Yoon, C. "Transition-Preserving State Tracking via Definition-Grounded Compilation." Manuscript under peer review, 2026. [GitHub](#)
- Yoon, C. "EvAudit: Latent Proxy-Evidence Auditing for Generative Data Mining." Manuscript under peer review, 2026. [GitHub](#) | [Software DOI](#)
- Yoon, C. "Evidence Is Not Free: Constrained Template Evidence Acquisition for Risk-Aware Governance." Manuscript under peer review, 2026. [GitHub](#) | [Software DOI](#)
- Yoon, C. "COLD-ACT: Low-Footprint Closed-Loop Active Vision for Cold-Chain Readiness Inspection." Manuscript under peer review, 2026.

Research in Development

- Yoon, C. "FigureSense: Provenance-Difference Verification for Chart-Generation Pipelines." Manuscript in preparation, 2026. Solo author.
- Yoon, C. "When Equivalent Equations Mimic Physical Change: Effect-Calibrated Semantic Selectivity in PDE Foundation Models." Research manuscript in progress, 2026. Solo author.

FACULTY-MENTORED RESEARCH EXPERIENCE

Independent Researcher - Scientific ML & Fractional PDEs

Sep. 2025-Jul. 2026

Faculty mentor: Kamran Kazmi, Professor of Mathematics, University of Wisconsin Oshkosh

- Designed and executed self-directed research cycles on fractional PDEs and Fourier neural operators, presenting research plans, code, results, and revisions for faculty critique.
- Developed QDC research on precision-sensitive failure in low-bit neural operators using Python/PyTorch, numerical PDE methods, residual analysis, controlled ablations, matrix-free LOBPCG, higher-resolution experiments, measured RealPDEBench data, and hardware-oriented evaluation.
- Studied numerical PDEs, error/convergence analysis, interpolation, and numerical differentiation/integration; continued the work beyond the credit-bearing independent study through the July 2026 WiSys SPARK Symposium.

Independent Researcher - Stochastic Processes & Vibration Modeling

Sep. 2025-May 2026

Faculty mentor: Ahmed O. Nasif, Associate Professor of Electrical Engineering Technology, University of Wisconsin Oshkosh

- Studied stochastic processes and probabilistic inequalities using selected material from Talagrand; independently implemented Python simulations, equation derivations, ODE and multi-DOF vibration models, Hamiltonian formulations, matrix/eigenvalue analysis, physics-constrained neural networks, and convergence/error analysis.
- Presented "Numerical Stability and Convergence of Physics-Constrained Neural Networks in Vibration Modeling" as an oral research talk at the 2025 Pi Mu Epsilon Undergraduate Regional Mathematics Conference.

SELECTED RESEARCH & ENGINEERING PROJECTS

CurbMap 360

Fall 2024-Summer 2025

- AI-assisted accessibility mapping prototype using autoencoder-, VAE-, and normalizing-flow-based anomaly/OOD screening for sidewalk scans; 1st Place, 2025 Culver Business Model Competition; later pitched at Fox Valley of the Pitch and the WiSys Big Idea Tournament. [GitHub](#)

MagTrace-MX: Software-Defined Quality Fixture

2026

- Developed an 8x8 inductive sensing work-surface concept with LC-tagged components, normalized feature maps, an embedded TinyML decoder, and a deterministic fail-safe workflow engine; 2026 Create the Future Design Contest entry. [GitHub](#)

DuctSense AFD

2026

- Developed a zero-penetration duct-diagnostics concept using structure-borne piezo sensing, an ambient reference channel, FFT-derived band features, and persistent local status logic; 2026 James Dyson Award entry. [GitHub](#)

SwallowSafe AI

2026

- Developed a clinician-programmed closed-loop assistive pacing concept combining throat-motion sensing, local edge processing, and an actuated cup mechanism for sense-compare-pace feedback; 2026 Create the Future Design Contest entry. [GitHub](#)

Assistive Hand-Gesture AI

Feb.-May 2025

- Developed a lightweight gesture-recognition prototype using hand-landmark geometry, temporal stabilization, pruning, and INT8-oriented inference optimization; presented through WiSys Quick Pitch activities. [GitHub](#)

RESEARCH PRESENTATIONS

- “Placement-Aware Quantization for Fractional-PDE Neural Surrogates.” WiSys SPARK Symposium, 2026. Poster presentation; faculty mentor: Kamran Kazmi. [DOI](#)
- “Numerical Stability and Convergence of Physics-Constrained Neural Networks in Vibration Modeling.” Pi Mu Epsilon Undergraduate Regional Mathematics Conference, 2025. Oral presentation. [DOI](#)

TECHNICAL SKILLS

Programming & ML: Python (4+ years); PyTorch; TensorFlow; scikit-learn; NumPy; pandas; SciPy; matplotlib.

Methods: scientific machine learning; Fourier/neural operators; numerical PDEs; quantization/QAT; residual correction; autoencoders; variational autoencoders; normalizing flows; anomaly/OOD detection; generative and unsupervised learning; uncertainty quantification; selective prediction; cross-validation; robustness and ablation design.

Research engineering: Git/GitHub; Zenodo archival releases; public research artifacts; LaTeX; experiment configuration; tests; result ledgers; reproducibility documentation; technical writing and presentation.

TEACHING, LEADERSHIP & SERVICE

- Center for Academic Resources Math Tutor, UW Oshkosh (Sep. 2025-May 2026); tutored 10-15 undergraduate students weekly in one-on-one and small-group sessions across calculus, college algebra, precalculus, and linear algebra.
- AI Club: Treasurer (Spring 2025-May 2026) and Vice President (Spring 2026); organized weekly paper/research discussions, workshops, and interactive AI/math activities.
- Computer Science Club: Treasurer (Spring 2025-May 2026). Social Justice Club: President (Sep. 2025-May 2026); previously coordinated finance and activity planning.

SELECTED HONORS & RECOGNITION

- Culver Business Model Competition - 1st Place, CurbMap 360 (2025); project featured by UW Oshkosh News and Universities of Wisconsin.
- Calculus Prize Exam Winner (2025).
- William Lowell Putnam Mathematical Competition competitor (2024).